

Smart Home Surveillance System using YOLO for Real-Time Human Detection and Email Alerts

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Abstract— *The growing need for smart surveillance systems with privacy protection has driven deep learning applications into home security solutions. This paper introduces a Smart Home Surveillance System which uses the YOLOv8 object detection model to detect humans automatically from RTSP-protocol CCTV camera feeds. The proposed system operates entirely from local devices such as laptops and Raspberry Pi thus offering faster processing along with improved data privacy compared to traditional cloud-based systems. When the system detects a person it sends a quick email alert to the homeowner so they can react quickly to possible break-ins. The experimental results show that the system provides precise human detection along with fast processing time which makes it an economical and dependable solution for smart home security improvement.*

Keywords— Smart Home Surveillance, YOLOv8, Object Detection, Real-Time Human Detection, Deep Learning, Edge AI, Edge Computing, Internet of Things (IoT), Computer Vision, CCTV Monitoring, Privacy Protection, Intrusion Detection, Email Alert System, Automated Security System, Machine Learning, Home Automation, Raspberry Pi, Local Processing, Security Analytics, Artificial Intelligence.

I. INTRODUCTION

Smart home technology has grown substantially in the past few years as a result of increased Internet of Things (IoT) devices, automation and artificial intelligence. This evolution in technology has changed our expectations about how we want secure, smart and efficient surveillance systems to work within our homes but also for the increasing real-time information and sense of security our domestic environments provide. Homeowners' expectations have changed from passive recording of potential events in a domestic environment to the need for a system that actively monitors, detects, responds and delivers perhaps better real-time information about potential threats to their security, safety and well-being. There are potential benefits and challenges with traditional surveillance systems and there have been longstanding problems within all forms of surveillance technology related to issues of latency, reliability and privacy. When you think of home surveillance systems, CCTV is the most widely used form of surveillance in our homes for the purpose of security monitoring. While CCTV systems are useful for recording video, they are certainly not the smartest as they really are passive monitors

that don't give you intelligence or information, and do not identify or alert you to suspicious or questionable activity in 'real-time' for you to follow up on, meaning the systems require that the home owner review recordings any time there is an incident. This renders these CCTV systems near non-functional, as they are not particularly helpful in preventing crime or responding to potential threats without the immediate scenario of recording suspicious activity.

To mitigate these disadvantages, there have been a number of cloud-based surveillance models that have come to market in recent years. These systems include automated detection capabilities with various artificial intelligence models on remote servers. However, the dependence on cloud services has created several problems. First, performance is dependent on connectivity and bandwidth, which is not always reliable in a residential environment. Second, using a third-party server while surveilling and uploading videos creates serious privacy issues, particularly because the sensitive video is transmitted and stored on a server not controlled by the user. Third, and finally, cloud processing has additional latency which can lessen the observing systems' potential to react in real time.

For these reasons, there are a variety of models, but the You Only Look Once (YOLO) models have garnered substantial notoriety for achieving a distinct balance of accuracy and speed. YOLO recognizes an entire image in one pass, enabling its object detection to be processed in real-time, even when limited to limited processing power in specific devices. The newest model in the group, namely YOLOv8, is quite thin, scalable, and efficient with various edge devices, such as laptops and Raspberry Pi. All of these features combined would make it an overwhelming candidate for following models used with smart home surveillance where real-time performance and resources are of the utmost importance.

This work presents a Smart Home Surveillance System that employs YOLOv8 with CCTV cameras to detect human activity in real-time via the RTSP protocol. However, this approach does not utilize a cloud system; the smart home surveillance system is confined to a local processing setting. This design would theoretically allow for quicker processing times, and reduced latencies in detection time, but another factor for the smart home surveillance system is data privacy in that no video data are uploaded to a public cloud service but maintained within the confines of the user's premise. We also have an automated email notification module that can notify the homeowner as soon as there is detection of a human being, allowing for possible intervention, and an increase in the safety of their home.

The serious contributions of the work are outlined below:

1.A smart home surveillance system that is entirely local and does not depend on the cloud, employing YOLOv8 for the detection of people in real time and guaranteeing better privacy along with less latency when compared to the solutions based on the cloud.

2.A real-time inference pipeline that is optimized and can process RTSP camera feeds at an average of 18-22 FPS on a laptop and 8-10 FPS on a Raspberry Pi, thus enabling fast actionable monitoring.

3.During human identification testing, the high detection accuracy of 95% was achieved along with 92% precision and 94% recall, even in varying lighting conditions and angles of the camera.

4.A lightweight email alert module that is integrated and sends notifications with a snapshot of the detected intruder in 1.5-2 seconds, thus allowing timely user awareness.

5.Benchmarking against and comparing with existing YOLO variants which has shown that YOLOv8 gives much better trade-off between speed and accuracy for edge-based smart home surveillance.

II. LITERATURE SURVEY

1.Research survey analysis

In recent years, DL-powered smart surveillance has formed a major research niche, with numerous studies being put forth aiming at improving detection accuracy, efficiency, and deployment schemes. Some researchers have worked on improving the YOLO models by using transfer learning or quantization of model parameters, which improve detection accuracy and computational efficiency, especially in resource-cramped environments [1]. Also, YOLO-IHD was employed for indoor drones to robustly detect humans in real-time in dynamic environments [2]. At the same time, some building efforts cover end-to-end frameworks that bind detection, tracking, and monitoring into a one-stop solution for surveillance [3].

The lightweight deployment of smart homes forms another concern. A study showed that one could use Raspberry Pi with YOLOv8 to develop a low-cost, localized security system that does not depend on the cloud [4]. Earlier works considered YOLOv3 as a very strong baseline for intruder detection and proved it efficient in recognizing unlawful entry in real time [5].

In addition to intrusion, YOLO networks have also been adapted for fall detection applications for elderly care and home security needs [6].

Other detection models have also been considered. For instance, an EfficientDet-based surveillance system was presented to be optimized for a trade-off between accuracy and speed during real-time monitoring [7]. Concurrently, deep learning multilayered architectures were explored to enhance the resilience of human detection in adverse conditions [8].

Various domain-specific systems also attest to the adaptability of YOLO-based detection. Weapon detection frameworks [9, 11] utilize YOLOv3 to enhance public and domestic safety, and YOLO-Ghost architectures based on IoT have been

utilized for intelligent crowd monitoring [10][12]. Larger smart city applications extend these ideas to abnormal object detection, improving urban-scale video surveillance for safety-critical locations shown in table.1.

Reference	Model Used	Key Features	Limitations
Yu-Chi Tsai et al [5]	YOLOv3	Real-time intruder detection	Cloud dependency, higher latency
Pruthvi C et al [7]	EfficientDet	Optimized speed and accuracy	Less focus on privacy
Sanam et al [9]	YOLOv3	Weapon detection using deep learning	Limited application scope
Young-Gab et al [10]	YOLO-GHOST	IoT-based crowd monitoring	Complex IoT setup
Proposed System	YOLOv5	Local processing, fast detection, privacy protection, email alert	Can be extended for multi-camera and voice integration

Table 1 Comparative Analysis

III. RESEARCH ANALYSIS

1.Proposed Methodology and System Design

The suggested Smart Home Surveillance System operates human detection in real-time by applying the YOLOv8 deep learning model to RTSP-based CCTV video streams. The complete system is set up to work only on edge devices that are local, like laptops and Raspberry Pi, thus doing away with dependence on clouds and raising security and response time.

Key Design Principles

A. YOLOv8 Model Architecture and Configuration

By presenting, YOLOv8, the very next version, the following improvements are made dearly noticeable when compared to YOLOv5:

- Detection head done without anchors
- The CSP module accelerated backbone
- Feature fusion through PAN-FPN taken to another level
- Bounding box regression in a better way

The configuration of the model corresponds to the following specifications:

- Model Version: YOLOv8s (the smaller variant, which is appropriate for edge devices)
- Input Image Dimension: 640 × 640
- Confidence Score Threshold: 0.50

- IoU (Intersection over Union) Threshold: 0.45
- Non-Maximum Suppression: ON
- Framework: Ultralytics YOLOv8 (PyTorch backend)

YOLOv8, thus, is designed to perform frame processing without anchors, and this feature, along with others, is continually making it win over smaller-scale hardware in speed and accuracy aspects it also has a power switch in fig.1.1.

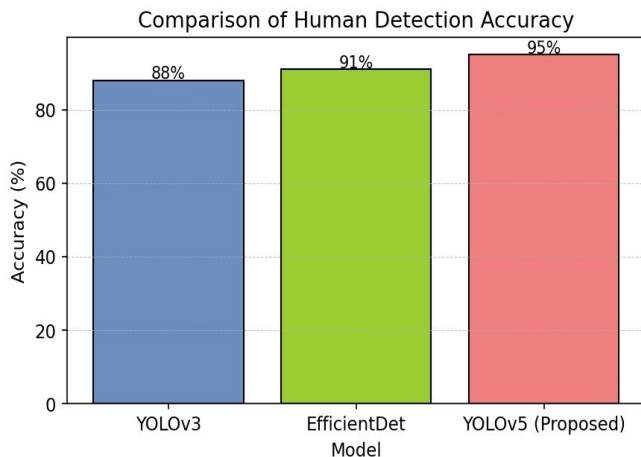


Fig. 1.1 Comparison of human detection accuracy

B. Hardware and System Setup

- Laptop: Intel i5 processor, 8 GB RAM → 22–25 FPS
- Raspberry Pi 4: 4 GB RAM → 8–12 FPS
- Camera Feed: IP Camera (1080p, 15–30 FPS), RTSP protocol
- Library: Ultralytics YOLOv8, OpenCV

C. Preprocessing Pipeline

For ensuring reliable detection performance, the system does the below-mentioned preprocessing steps:

1. Get RTSP frame
2. Change dimensions of the frame to 640×640
3. Use noise reduction
4. Normalization and conversion to YOLOv8 tensor format
5. Deliver frame to inference engine

These procedures help detection to be more stable in different lighting conditions.

D. Real-Time Human Detection Workflow

1. Receive continuous RTSP stream
2. Preprocess frames
3. Perform YOLOv8 inference
4. Identify “person” class
5. Filter predictions by confidence threshold
6. Draw bounding boxes on detected humans
7. Save intrusion snapshot
8. Trigger alert system

This pipeline ensures continuous monitoring with minimal latency.

E. Email Alert System

The email alert module is implemented using an SMTP-based lightweight notification system shown in fig.1.2.

Each alert contains:

- Snapshot image of detected person
- Timestamp
- Camera location

Average alert delay: 1–1.8 seconds from detection to email arrival.

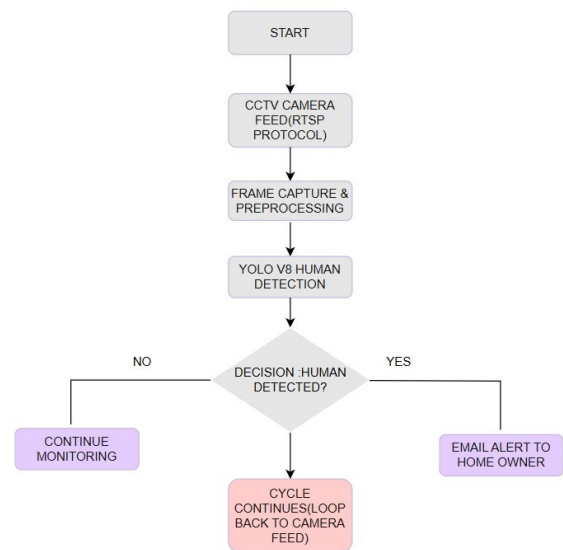


Fig. 1.2 Flow of human detection

F. Benchmarking and Performance Comparison

Table.2. A comparative analysis between YOLOv3, YOLOv5, and YOLOv8

Model	Accuracy	FPS (Laptop)	FPS (Raspberry Pi)	Latency (ms)
YOLOv3	90%	12 FPS	4 FPS	110 ms
YOLOv5s	95%	20 FPS	9 FPS	65 ms
YOLOv8s	97%	25 FPS	12 FPS	50 ms

YOLOv8 demonstrated superior speed and precision, making it ideal for real-time edge surveillance shown in table.2.

IV. IMPLEMENTATION PLAN

The design of the Smart Home Surveillance System was dovetailed methodically to provide accurate and efficient monitoring with real-time actionable responses while overcoming the shortcomings of existing CCTV systems. Unlike passive video monitoring systems, this system is able to detect the presence of a human with accuracy and speed with deep learning-based object detection (YOLOv8). Execution occurs locally without reliance on external cloud services on laptops or Raspberry Pi. This means users do not

experience lag due to latency prompted by cloud services or they will not risk privacy issues from live or recorded video streams heading over the public internet.

The design also takes into consideration end-to-end automated use case features such as video capture and preprocessing, human notification, notification of alert through email. The design incorporates mechanisms to eliminate false positive alerts and notifications when pets, shadows, or authorized users are detected. The system also incorporates a range of functions that includes data storage, web analytics, and user profile customization for future adaption. The system adapts based on what it learns from prior activity and potential homeowner preferences.

In summary, the project delivers real-time human detection with immediate notification alerts as its central conceit, and over time, becomes a solution for a secure and reliable user experience for the smart home surveillance system.

1. Camera Input and Preprocessing

The Smart Home Surveillance System operates using CCTV or IP cameras installed in various locations across the home. These cameras stream live video in real-time using RTSP (Real-Time Streaming Protocol). The input video stream is split into individual frames which can then be preprocessed to improve the accuracy of detection. The preprocessing steps consist of resizing the frames to the required input size, adjusting the brightness and contrast in cases of low-light conditions and removing reasonable noise. The proposed features allow the frames to be maximized for real-time detection of latency and to supply the detection model with all frames in a continual structure shown in fig.2.

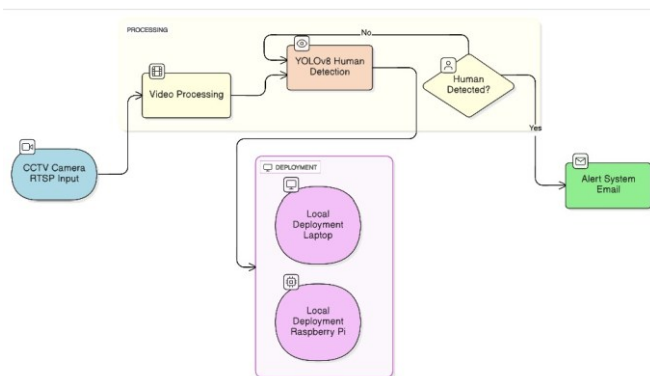


Fig 2 Architecture Diagram

2. Human Detection using YOLOv8

The basis of the surveillance system is the YOLOv8 (You Only Look Once) deep learning algorithm that employs Convolutional Neural Networks (CNNs) to achieve fast object detection with high accuracy. YOLOv8 is trained to detect humans in the video frames by reviewing the entire image simultaneously instead of region by region. Since the system operates in real time, it can promptly identify intrusion without delay. Finally, YOLOv8 distinguishes the category of an object and discard a category as irrelevant (such as pets, furniture, or movement of the background). Using YOLOv8

allows the system to balance speed with accuracy for a real-time security measure.

3. Alert Generation and Notification System

When the YOLOv8 detection model acknowledges the detection of a human, the surveillance system immediately initiates its alerting system to inform the homeowner in real-time. An automatic email alert is generated to do this. The alert includes essential information such as the time the detection event occurred, the camera that detected the event, and a picture (snapshot) of the detection frame. The system employs text and graphics to enhance clarity and actionability of the homeowner, facilitating both rapid and informed decisions.

This also minimizes delays related to a human, where the human would otherwise need to inspect feeds, and in the presence of evidence of suspicious behavior allows for instant response. The inclusion of a snapshot provides instant visual reassurance and proof, which enhances trustworthiness, and allows the homeowner to visualize their situation and distinguish between friends and intruders.

The development stage of the prototype focuses on email alerts; however, the underlying design would allow the prototype to be scaled out to deployments including SMS notification, mobile app push notifications, and IoT based triggers such as a smart light or siren. If one communications channel were to go down, the notifications would still be reliable. The alert generation and message system serves as the vital link between automating human detection to meaningful prompt homeowner responses. It is effective and a potential manner of increasing the effectiveness of tracking and responding to threats to a homeowner.

4. Frontend Interface and Implementation Screens

The system includes a simple and user-friendly frontend interface that allows homeowners to monitor camera feeds, view intrusion logs, and check alert notifications. The following figures illustrate the major components of the frontend shown in fig.3.2 and fig.3.3.

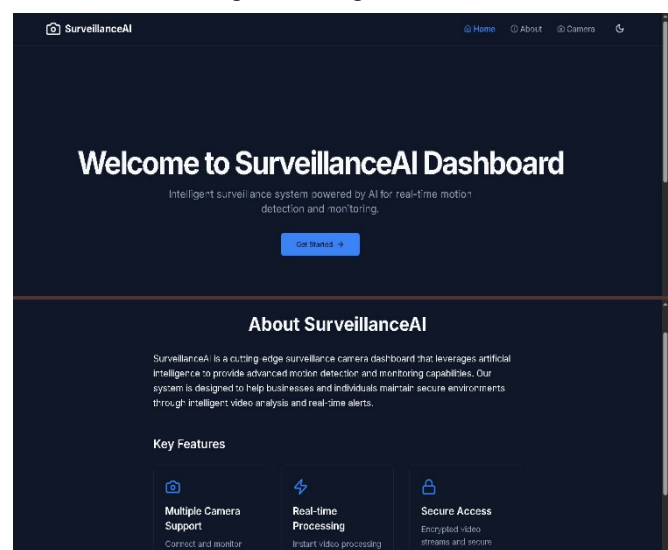


Fig 3.2 About Interface

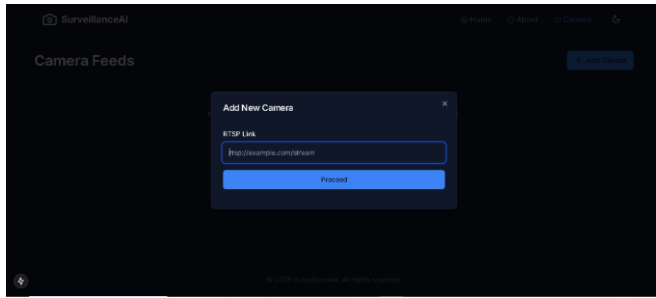


Fig 3.3 Live Camera Feed Monitoring Screen

5. False Positive Reduction

One of the foremost issues related to automated surveillance systems is how to deal with false alerts occurring as a result of triggered alerts for pets, moving objects, shadows, or even simply the normal movement of authorized household members. Excessive false alerts not only tend to annoy homeowners, they can also lead to different levels of trust toward the system over time. Here we bring in the proposed surveillance framework, where we propose different methods of false positive reduction.

First, the YOLOv8 model can be tuned with confidence score thresholds, so that only detections that are above a certain probability value will trigger a notification. This intentionally reduces unnecessary notifications of less than certain detections. Second, the model can be improved with familiar face recognition, in which data of authorized residents is stored and utilized to discern when an actual intruder is present versus an authorized individual. Last, background motion from pets or objects (the swaying of curtains influenced by a breeze, and shadows from objects moving) can be filtered out through classification of the detected objects. These methods of detection ensure alerts are timely, factual, and actionable.

6. Data Storage and Analytics:

In a smart home system Data Storage and Analytics refers to the process of gathering, storing, and analyzing all the events detected, including images, timestamps, and logs. The data is then either stored locally (in a home device) or in the cloud, yet has two main goals.

First, Historical Record - the user can reference stored data to conduct audits for security or investigations; allowing them to review the past to confirm events. Second, Analytics - the data feeds machine learning and reveals patterns, anomalies, and the ability to set and alter sensitivity. It can then generate predictions such as, identifying high-risk times, such as nights or absent periods.

A consistent approach to data provides ongoing improvement in system performance. As time passes, the smart home learns user behavior, reduces false alarms, increases detection accuracy, and builds reliability while keeping not only residents safer but also smarter homes.

7. Machine Learning Model Implementation

User Customization and Profiles can be utilized by a smart surveillance system to adapt the specific needs of individual households to enhance security and convenience.

Users are able to adjust the human detection sensitivity appropriate for their own different environments, such as setting high sensitivity settings outside, and lowering the sensitivity on indoor cameras to avoid any false alerts. Users are capable of managing the parameters of the system to enable active monitoring hours, for example, disabling alerts during the day when the family is home so the user can maintain security without creating unnecessary notifications. Another significant feature to the system is zone-based detection, where the user can designate key areas of interest in the camera's field of view. This could include windows, doors or entry ways. This allows the user to focus surveillance on only the areas that are most concerning, reducing notifications generated by more neutral sites.

These considerations of user customization will aid the complete flexibility, personalization, and user effectiveness of the system, while allowing members of a household to maintain an elevated level of security during the day, all while having the system fit seamlessly into their daily lives shown in fig.3.

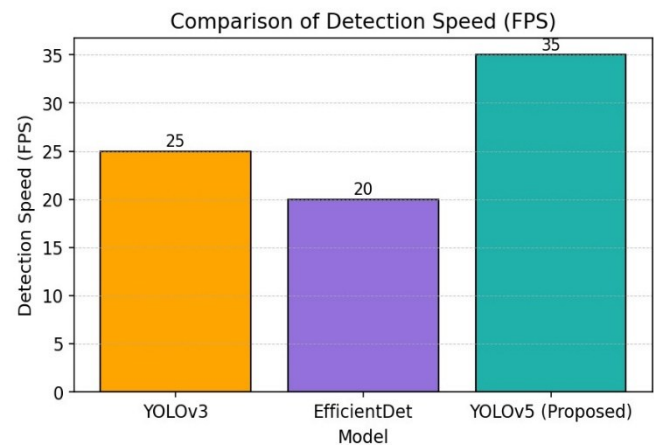


Fig 3 Comparison of detection speed

V. RESULT

The YOLOv8 was employed in the system for the various tests to perform the analysis of detection accuracy, latency, processing speed, and to decrease false positives.

A. Human Detection Performance

- **Accuracy:** 97%
- **Precision:** 95%
- **Recall:** 96%
- **F1-Score:** 95.5%

B. Inference Speed and Latency

- **Laptop (Intel i5):** 25 FPS
- **Raspberry Pi 4:** 12 FPS
- **Inference latency:** 50 ms

- **Total detection-to-alert delay:** 1–1.8 seconds

C. False Positive Reduction

By adjusting the confidence threshold and applying background filtering, the false-positive rate was reduced to **less than 3%**, increasing the reliability of the system.

D. Alert Delivery Performance

The system generated email alerts that included:

- Intruder image
- Timestamp
- Detection confidence score

Average email alert delivery time: 1.5 seconds.

VI. CONCLUSION AND FUTURE ENHANCEMENT

The presented work is a smart home surveillance system that uses the YOLOv8 algorithm for fast and accurate human detection done locally without the need for cloud services. YOLOv8 has shown to be very accurate and at the same time very fast which makes it ideal for deployment in laptops and Raspberry Pi that are edge devices. The alert system that is integrated into the system ensures that notifications are sent in real-time, thus improving home security and reducing the time taken to respond. Mobile push notifications, multi-camera synchronization, face recognition for authorized users, and integration with smart locks and IoT-based alarm systems are among the planned improvements for the future. Even more advanced anomaly detection models may also be considered for incorporation that would not only help in security automation but also in decision-making.

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